Do Now · retrieval

Lesson 119 · Spiral Development and Method Selection

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is one distinguishing XP practice?

Q2

An OR gate has inputs 0 and 1. What is the output?

Q3

What is the effect of enqueue(X), enqueue(Y), dequeue() on an initially empty FIFO queue?

Q4

What is the time complexity of sequential search in the worst case?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Plan the cycle — Set objectives, options and constraints for a spiral iteration.

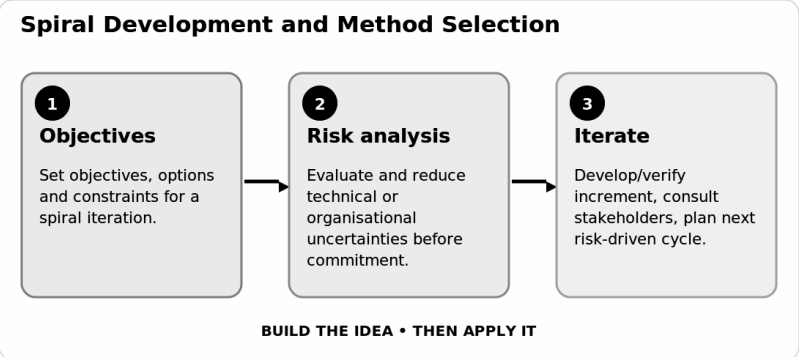
Core idea 2

Analyse risks — Evaluate and reduce technical or organisational uncertainties before commitment.

Core idea 3

Prototype and review — Develop/verify increment, consult stakeholders, plan next risk-driven cycle.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

spiral model — Risk-driven iterative development lifecycle.

risk analysis — Estimating likelihood and impact of potential problems.

prototype — Early model built to explore or test an idea.

iteration — Repeated cycle of development and evaluation.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

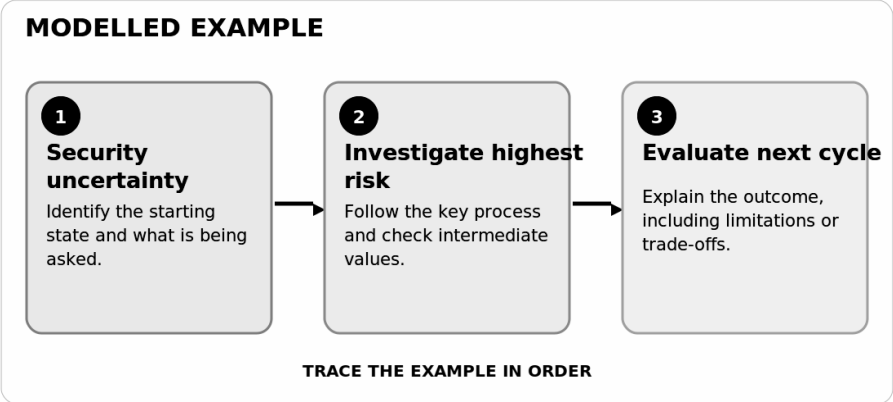
Problem

A bank is building a novel payment verification system with security uncertainty. Justify the spiral model.

Follow the worked method

1. Model answer — Each cycle sets goals, identifies risks such as fraud or performance, prototypes/assesses the riskiest elements, evaluates with stakeholders then plans the next cycle. Risk reduction drives the project. — Check: link each claim to the scenario and use precise OCR terminology.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Describe three central activities within one spiral development cycle. [6]

Question 2

Give two reasons a fixed-requirement small project might not benefit from the spiral model. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare waterfall, agile/RAD, XP and spiral for a high-risk changing specification. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Spiral development is simply waterfall repeated without evaluating project risks.

Use key vocabulary from memory

1. Explain spiral model in your own words.

2. Explain risk analysis in your own words.

3. Explain prototype in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.